

Anant Khanna

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EDUCATION

University of British Columbia

Bachelor of Applied Science in Computer Engineering - Dean's List 2025

Vancouver, BC

Expected Graduation: 2028

EXPERIENCE

Founding Software Engineer | Agriprix

May 2026 – Present

- Rebuilt the hybrid intent router (deterministic rules + **LLM** classification) for an agent platform spanning **70+** tools – reaching **96%** routing accuracy, **45%** lower median latency, and **25%** lower per-request token cost – with graceful fallback to deterministic rules on classifier timeout and per-call token/latency observability
- Implemented **Stripe** authorize-and-capture checkout in a **Django/DRF** backend deployed via **Docker**, with deterministic idempotency keys, an integer-cent commission pricing engine, and per-farm fulfillment state machines for multi-supplier orders
- Led a platform security audit surfacing **67** vulnerabilities (5 critical): patched mass-assignment/IDOR flaws in Django REST Framework serializers, eliminated overselling races via row-level locking, and shipped a 15-step end-to-end Playwright journey covering the supplier-to-buyer order flow
- Shipped a WhatsApp AI commerce assistant on Meta's Business Cloud API: buyers place real orders over chat through native Confirm/Cancel buttons on the agent's confirmation rails, with HMAC-verified webhooks and idempotent double-tap protection

Machine Learning Software Engineer Team Lead | UBC UAS

Sept. 2024 – Present

- Integrated YOLOv11 (**PyTorch**) pipeline in Python and C++, enabling real-time autonomous object detection on embedded systems travelling at **50+** km/h, optimizing inference for low-latency execution
- Implemented **TensorRT** FP16 quantization to compress model weights, reducing memory bandwidth usage by **50%** while maintaining inference precision within **0.5%** of the baseline
- Achieved **95%** detection accuracy with model optimization, improving mAP by **22%**

Co-founder & Lead Developer | UBC Finds

Oct. 2025 – May 2026

- Built a campus utility finder map on **React/Next.js** and Supabase, architecting event-driven state synchronization over Supabase realtime channels that eliminates race conditions through atomic writes

PROJECTS

GPU Inference Benchmark | Python, PyTorch, CUDA/TensorRT, ROCm/MIGraphX

July – Aug. 2026

- Built a three-rung ladder (**PyTorch**, **TensorRT** FP32, **TensorRT** FP16) isolating compilation vs. precision gains
- Ported the harness to AMD MI300X via ROCm/MIGraphX: unoptimized **PyTorch** FP32 there (**1,143** fps) beats the fully-optimized **TensorRT** FP16 engine on an RTX 3080 (**1,008** fps)
- Scaled MIGraphX FP16 batch 1 to 8 with near-zero latency penalty (**1.1ms**/inference), reaching **7,277** fps

Wealth Council | Python, FastAPI, React, Google Gemini, Fetch.ai

Mar. 2026

- Engineered an agentic investment platform, winning an award at the ProduHacks x Fetch.ai hackathon
- Architected a **Multi-Agent Orchestration** Layer using Fetch.ai uAgents to asynchronously compute portfolio risk metrics and integrated FinBERT (NLP) to perform sentiment analysis on live financial news streams

GuardCam: AI Dashcam | Python, React Native, WebSockets, OpenCV

Mar. 2026

- Developed an edge-to-cloud driver monitoring system to detect micro-sleeps, utilizing a custom TCP WebSocket bridge to bypass HTTP overhead for low-latency video streaming
- Implemented MediaPipe and OpenCV to map a **468**-point 3D face-mesh, calculating Eye Aspect Ratio (EAR) to trigger sub-second haptic JSON alerts

FPGA-Based Neural Network Inference Accelerator | Verilog, FPGA, C++, ARM

July 2025

- Designed and implemented a single-layer neural network (MAC unit) on a DE1-SoC **FPGA** using **Verilog**
- Achieved **3,000** inferences/sec with a **300%** performance speedup over ARM-only execution by optimizing HPS-FPGA communication

TECHNICAL SKILLS

Languages: Python, C, C++, C#, SQL, Java, JavaScript, TypeScript, RISC-V & ARM Assembly

AI/ML: LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, Generative AI, Conversational AI, Transformers, AI Evaluation, Intent Recognition

Frameworks: Django/DRF, FastAPI, Flask, Celery, React, Next.js, PyTorch, TensorFlow, YOLO/Ultralytics, NumPy, Matplotlib, REST APIs

Tools & Cloud: Google Cloud Platform, Docker, PostgreSQL, MySQL, Supabase, Stripe, Playwright, Git, Linux

Hardware/RTL: FPGA Design (Intel Quartus, QuestaSim, Platform Designer), Verilog, SystemVerilog, ARM SoCs